Definition 3.14: Let $f:\mathbb{R}^n \to \mathbb{R}$ be C^2, so that all of its 2^{nd} order partial derivatives commute. The symmetric matrix

$$\text{Hess} f(x) = D^2 f(x) = \begin{bmatrix} f_{11} & \cdots & f_{1n} \\ \vdots & \ddots & \vdots \\ f_{n1} & \cdots & f_{nn} \end{bmatrix}$$

is called the Hessian of $f(x)$. This matrix acts as a bilinear form, in a sense to be made clear in Definition 3.17, which is the reason for the two lowered indices.

Our interest in the Hessian here is due to it's role in identifying critical points of functions, as well as it's appearance in the multivariate Taylor Theorem:

Application I: Taylor's Theorem in n-dimensions

Lets quickly review Taylor's Theorem in 1-dimension:

Let $f:(a,b) \to \mathbb{R}$ be a C^k function for some $k \geq 0$. Fix a point $x_0 \in (a,b)$, and suppose we want to approximate $f(x)$ for x near to x_0 by much simpler functions: polynomials.

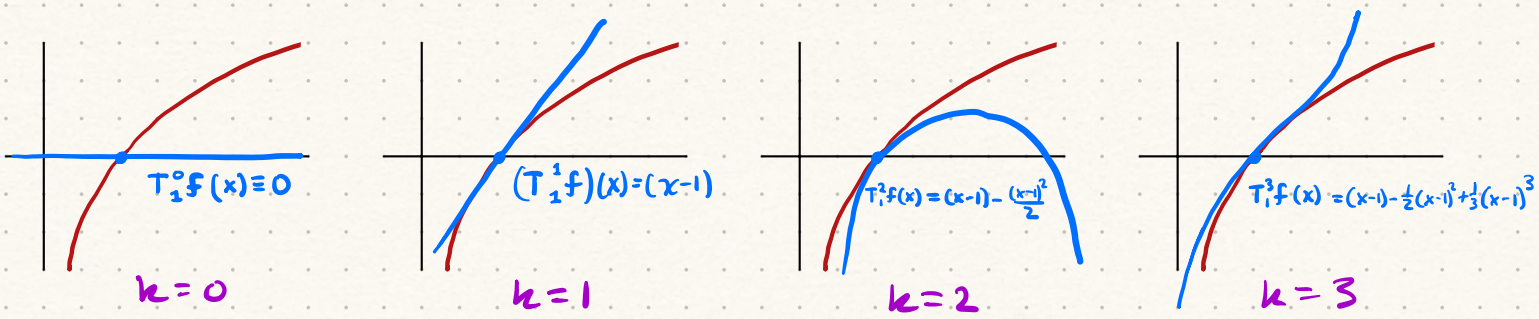
Fact: If $p:(a,b) \to \mathbb{R}$ is a degree $k \geq 0$ polynomial that agrees with $f:(a,b) \to \mathbb{R}$ up to k^{th} order at x_0, meaning

$$p(x_0) = f(x_0), \quad p'(x_0) = f'(x_0), \quad \ldots, \quad p^{(k)}(x_0) = f^{(k)}(x_0),$$

then $$p(x) = f(x_0) + f'(x_0)\cdot(x-x_0) + \cdots + \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k.$$

Definition: We call this $p(x) \equiv T^k_{x_0} f(x)$ the k^{th}-order Taylor polynomial of $f(x)$ at x_0.

Pictorially, we have the following e.g. with $f(x) = \ln x$ and $x_0 = 1$:



The next result captures how well these polynomials actually approximate $f(x)$ near x_0:

Taylor's Theorem 3.15: Let $f:(a,b) \to \mathbb{R}$ be C^k and fix $x_0 \in (a,b)$. Then the error function $R^k_{x_0}(x) = f(x) - T^k_{x_0} f(x)$ is C^k, and it satisfies

$$\lim_{x \to x_0} \frac{R^k_{x_0}(x)}{|x - x_0|^k} = 0.$$

Remark: There are several different ways to characterize and estimate $R^k_{x_0}(x)$. We won't prove them here, but you can easily find such results in the literature. For example, one can show that there is always a $c \in (a,b)$ between x_0 and x so that $R^k_{x_0}(x) = \frac{f^{(k+1)}(c)}{(k+1)!}(x - x_0)^{k+1}$, in the case where f is C^{k+1}. This can be proven using the Mean Value Theorem. In particular, this shows that $R^{k+1}_{x_0}(x) = f(x) - T^{k+1}_{x_0} f(x) = \frac{(x-x_0)^{k+1}}{(k+1)!}\left(f^{(k+1)}(c) - f^{(k+1)}(x_0)\right)$.

We want to generalize Theorem 3.15 to functions $f: U \subseteq \mathbb{R}^n \to \mathbb{R}$. Fortunately, this actually ends up being pretty easy (for real!) - the most difficult hurdle is just the notation:

Multivariate Taylor's Theorem 3.16: Let $f: U \subseteq \mathbb{R}^n \to \mathbb{R}$ be C^k at a point $x_0 \in U$ for some $k \in \mathbb{Z}^{\geq 0}$, and set.

$$T^k_{x_0} f(x) := \sum_{|\alpha| \leq k} \frac{D^\alpha f(x_0)}{\alpha!} \cdot (x - x_0)^\alpha$$

$$= f(x_0) + \sum_{i=1}^n D_i f(x_0)(x^i - x_0^i) + \frac{1}{2}\sum_{i,j=1}^n D_{ij} f(x_0)(x^i - x_0^i)(x^j - x_0^j) + \cdots$$

$$= f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2}(x - x_0)^t \cdot D^2 f(x_0) \cdot (x - x_0) + \cdots$$

Then the C^k function $R^k_{x_0}(x) := f(x) - T^k_{x_0} f(x)$ has

$$\lim_{x \to x_0} \frac{R^k_{x_0}(x)}{\|x - x_0\|^k} = 0 .$$

In general, this means that as $x \to x_0$, the behavior of $f(x)$ is the "same" as the polynomial $T^k_{x_0} f(x)$ up to order k, and that the error is in this approximation is <u>smaller</u>, on the order of $(k+1)$. We'll use this observation below when we talk about critical points.

<u>Proof of the Multivariable Taylor Theorem</u>: There are only two ingredients: the <u>1-variable Taylor Theorem</u> and the <u>Chain Rule</u>. Since the conclusion of the theorem, namely the behavior of $R^k_{x_0}(x)$ near x_0, only depends upon x near x_0, we can begin by assuming that $x \in B_\varepsilon(x_0) \subseteq U$, so that $x = x_0 + h$ for some $|h| < \varepsilon$ and the entire line segment $\gamma(t) = x_0 + th$, $t \in [-1,1]$, sits inside U. We then consider the 1-variable C^k function

$$f(\gamma(t)) : [-1,1] \longrightarrow \mathbb{R}$$

which has $f(\gamma(0)) = f(x_0)$, $f(\gamma(1)) = f(x)$, and

$$(f \circ \gamma)(t) = f \circ \gamma(0) + (f\circ\gamma)'(0)\,t + \cdots + \frac{(f\circ\gamma)^{(k)}(0)}{k!}\,t^k + R^{k,h}_0(\gamma(t))$$

where we emphasize that $R^{k,h}_0(\gamma(t))$ depends on h. By the chain rule:

- $(f\circ\gamma)'(t) = f'(\gamma(t))\cdot\gamma'(t) \Rightarrow (f\circ\gamma)'(0) = f'(x_0)\cdot h = \sum_{i=1}^n D_i f(x_0)\,h^i$
- $(f\circ\gamma)''(t) = [(f'\circ\gamma)\cdot\gamma']'(t) = \left[\sum_{i=1}^n (D_i f)(\gamma(t))\cdot h^i\right]'(t) = \sum_{i,j=1}^n D_{ij} f(\gamma(t))\,h^i h^j$

  $\vdots$

  $\Rightarrow (f\circ\gamma)''(0) = D^2 f(x_0)(h,h) = h^t \cdot D^2 f(x_0)\cdot h$

- $(f\circ\gamma)^{(k)}(0) = \cdots = \sum_{i_1,\ldots,i_k=1}^n D_{i_1 i_2 \cdots i_k} f(x_0)\,h^{i_1} h^{i_2}\cdots h^{i_k}$

and thus, evaluating at $t = 1$, we get (noting $h = x - x_0$)

$$\begin{aligned} f(x) &= f(x_0) + \sum_{i=1}^n (D_i f)(x_0)(x^i - x_0^i) + \tfrac{1}{2}\sum_{i,j=1}^n D_{ij} f(x_0)(x^i - x_0^i)(x^j - x_0^j) + \cdots \\ &\quad + \frac{1}{k!}\sum_{i_1,\ldots,i_k=1}^n D_{i_1 i_2 \cdots i_k} f(x_0)(x^{i_1} - x_0^{i_1})\cdots(x^{i_k} - x_0^{i_k}) + R^{k,h}_0(x_0 + th) \\ &\equiv (T^k_{x_0} f)(x) + R^k_{x_0}(x) . \end{aligned}$$

Notice that we pass from $\frac{1}{j!}\sum_{i_1,\dots,i_j=1}^{n} D_{i_1\cdots i_j} f(x_0)(x^{i_1}-x_0^{i_1})\cdots(x^{i_j}-x_0^{i_j})$ to the term $\sum_{|\alpha|=j} \frac{D^\alpha f(x_0)}{\alpha!}(x-x_0)^\alpha$ in $P_k(x)$ using Clairaut's Theorem: there are several ways to compute the derivative $D^{(\alpha_1,\dots,\alpha_n)} f(x_0)$ with $|\alpha|=j$: there are (α_i)-D_i derivatives and they can be taken in any order for a total of j derivatives. Thus, there are $\binom{j}{\alpha_1,\dots,\alpha_n} = \frac{j!}{\alpha_1!\cdots\alpha_n!} = \frac{j!}{\alpha!}$ ways to compute $D^\alpha f(x_0)$ in the form $D_{i_1\cdots i_j} f(x_0)$, with $i_\ell = 1$ α_1-times, $i_\ell = 2$ α_2-times, ... etc: i.e. $\sum_{\left\{\substack{i_1,\dots,i_j \text{ such that} \\ D_{i_1\cdots i_j} f(x_0) = D^\alpha f(x_0)}\right\}} D_{i_1\cdots i_j} f(x_0) = \frac{j!}{\alpha!} D^\alpha f(x_0)$.

In any case, the point of making this computation is to show that in the multivariate case, $R^k_{x_0}(x)$ comes from the remainder term $R^{k,h}_0(\underbrace{x_0+th}_{\gamma(t)})$ in the 1-D Taylor Theorem. We know that

$$\lim_{t\to 0} \frac{|R^{k,h}_0(x_0+th)|}{\|th\|^k} = 0$$

for each fixed $h\in\mathbb{R}^n$, $\|h\|<\varepsilon$, and we must show that in fact

$$\lim_{x\to x_0} \frac{R^k_{x_0}(x)}{\|x-x_0\|^k} = 0,$$

To this end, we recall one of the particular forms of the remainder from the 1D-Taylor Theorem from above: for some $0\le t^* \le t$

$$R^k_{x_0}(x) = R^{k,h}_0(x_0+th) = \frac{\|th\|^k}{k!}\left((f\circ\gamma)^{(k)}(t^*) - (f\circ\gamma)^{(k)}(0)\right)$$

$$= \frac{\|th\|^k}{k!}\left[\sum_{i_1,\dots,i_k=1}^{n} (D_{i_1\cdots i_k} f)(x_0+t^*h) - D_{i_1\cdots i_k} f)(x_0)\right]$$

hence

$$\frac{R^k_{x_0}(x)}{\|x-x_0\|^k} = \frac{R^{k,h}_0(x_0+th)}{\|th\|^k} = \frac{1}{k!}\left[\sum_{i_1,\dots,i_k=1}^{n} (D_{i_1\cdots i_k} f)(x_0+t^*h) - D_{i_1\cdots i_k} f)(x_0)\right].$$

where we reiterate that this formula holds for any $x\in B_\varepsilon(x_0)$, $x = x_0+th$ for some $\|h\|<\varepsilon$ and $t\in[-1,1]$. Using that $f\in C^k \Rightarrow D_{i_1\cdots i_k} f : U\to\mathbb{R}$ are continuous, hence uniformly continuous on $\overline{B_\varepsilon(x)}\subseteq U$, we see that if $x\to x_0$, $\|t^*h\|\le\|th\|\to 0$, then the RHS above must $\to 0$ as well.

Examples/Exercises: Find $T^0_{x_0}, T^1_{x_0}, T^2_{x_0}$ for the following functions

1) $f(x,y) = e^{xy}$, $(x_0,y_0)=(1,1)$

2) $g(x,y,z) = \sin(x+yz)$, $(x_0,y_0,z_0)=(0,0,0)$

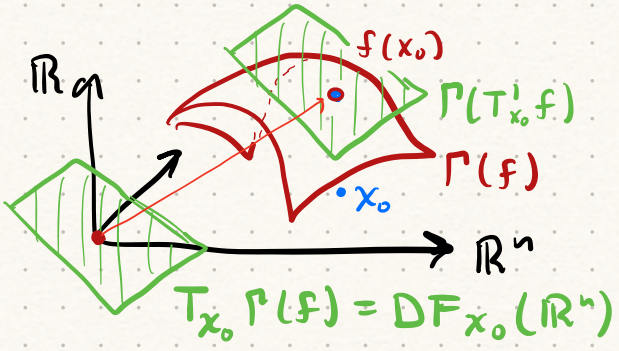
3) $h(x,y,z) = x^2+y^2+z^2$, $(x_0,y_0,z_0)=(1,1,1)$

Some Geometric Remarks:

In 1D we can draw attractive pictures illustrating how the graphs of the Taylor polynomials $T^k_{x_0} f(x)$ become better approximations to the graph of $f(x)$ near x_0 as $k \to \infty$. In higher dimensions we can obtain the same sort of geometric picture, even if it becomes hard to visualize in higher dimensions.

Let $f: \mathbb{R}^n \to \mathbb{R}$ and consider $\Gamma(f) = \text{graph}(f) = \{(x, f(x)) \in \mathbb{R}^n \times \mathbb{R}\}$ (it may help to think explicitly in 2D here, so the $\Gamma(f)$ is a 2D object). Suppose $f \in C^\infty$ for simplicity, and let $x_0 \in \mathbb{R}^n$.

- At 1st order, $T^1_{x_0} f(x) = f(x_0) + Df_{x_0}(x - x_0)$ has $\Gamma(T^1_{x_0} f) = n$-dim'l plane through $f(x_0)$, and is the best plane which approximates $\Gamma(f)$ near x_0. If we consider the graph map $F(x) = (x, f(x))$ (which maps $\mathbb{R}^n$ to $\Gamma(f)$ in $\mathbb{R}^{n+1}$), then $DF_{x_0}: \mathbb{R}^n \to \mathbb{R}^{n+1}$ has $DF_{x_0}(\mathbb{R}^n) := T_{x_0}\Gamma(f)$ an n-dim'l subspace of $\mathbb{R}^{n+1}$, called the **tangent plane** to $\Gamma(f)$ at x_0. In particular, $\Gamma(T^1_{x_0} f) = f(x_0) + DF_{x_0}(\mathbb{R}^n)$



- At 2nd order, $T^2_{x_0} f(x) = f(x_0) + Df_{x_0}(x - x_0) + \frac{1}{2} D^2 f_{x_0}(x - x_0, x - x_0)$ (see Definition 3.17 below for the details of how $D^2 f_{x_0}$ acts as a quadratic form) has $\Gamma(T^2_{x_0} f)$ a n-dim'l surface in $\mathbb{R}^{n+1}$ through $f(x_0)$, and is the best quadric surface approximating the graph $\Gamma(f)$ near x_0. See the following section for more about this!

Bonus Section: A Linear Algebraic Perspective on Taylor's Theorem

There is a really interesting way to view Taylor's Theorem as a result about a direct sum decomposition of a space of functions. To set the stage, we introduce the following definitions. Let $\Omega \subseteq \mathbb{R}^n$ be an open domain and $x_0 \in \Omega$.

Definition A: Let $\sim$ be the equivalence relation on $C^k(\Omega;\mathbb{R})$ defined by setting $f \sim g$ iff $D^\alpha f(x_0) = D^\alpha g(x_0)$ $\forall \alpha$ with $|\alpha| \leq k$. Then define the k^{th} jet space at x_0 by $J^k_{x_0}(\Omega) := C^k(\Omega)/\sim$.

For each $f \in C^k(\Omega)$, its equivalence class $[f] \in J^k_{x_0}(\Omega)$ is called the k-jet of f at x_0, and this equivalence class is uniquely associated to the vector $J^k_{x_0}(f) := (D^\alpha f(x_0))_{|\alpha| \leq k} \in \mathbb{R}^N$ where $N = \binom{n+k}{n}$.

Thus, $J^k_{x_0}(\Omega) \approx \mathbb{R}^N$ under the isomorphism $[f] \longleftrightarrow J^k_{x_0}(f)$. The kernel of this map is exactly $\mathcal{I}^{k+1}_{x_0}(\Omega) := \{f \in C^k(\Omega) : D^\alpha f(x_0) = 0 \ \forall |\alpha| \leq k\}$.

Thus we have a short exact sequence (☨ See end of section!)

$$0 \longrightarrow \mathcal{I}^{k+1}_{x_0}(\Omega) \longrightarrow C^k(\Omega) \xrightarrow{J^k_{x_0}(-)} \mathbb{R}^N \longrightarrow 0$$

and hence $C^k(\Omega) \approx J^k_{x_0}(\Omega) \oplus \mathcal{I}^{k+1}_{x_0}(\Omega)$, or $J^k_{x_0}(\Omega) \approx C^k(\Omega)/\mathcal{I}^{k+1}_{x_0}(\Omega)$.

The content of this last decomposition is just a simple remark: a function $f \in C^k(\Omega;\mathbb{R})$ can be uniquely written as two pieces: one containing only the differential data of f at x_0 up to and including order k, and the "error" term which contains everything else (itself having vanishing data up to order k at x_0). The lower order part has a nice description (cf. the earlier Fact above):

Definition B: Let $\mathcal{P}^k(\Omega) := \{\text{polynomials of degree} \leq k \text{ on } \Omega\} \subseteq C^k(\Omega)$. Then $\mathcal{P}^k(\Omega) \approx \mathbb{R}^N \approx J^k_{x_0}(\Omega)$, since a polynomial of degree $\leq k$ is uniquely determined by its k-jet at a point: If $p \in \mathcal{P}^k(\Omega)$, we can write $p(x) = \sum_{|\alpha| \leq k} p_\alpha \cdot (x - x_0)^\alpha$ and it follows that we must have $p_\alpha = \frac{D^\alpha p(x_0)}{\alpha!}$.

Since this lower order piece of the direct sum decomposition is finite dimensional, it's easy to give it a nice inner product:

Definition C: Define an inner product on k-jets at x_0 by setting

$$\underbrace{\langle f, g\rangle_{J^k_{x_0}(\Omega)}}_{\text{on } C^k(\Omega)\text{ (not definite here!)}} = \underbrace{\langle [f], [g]\rangle_{J^k_{x_0}(\Omega)}}_{\text{on } J^k_{x_0}(\Omega)} = \underbrace{\langle J^k_{x_0}(f), J^k_{x_0}(g)\rangle_{J^k_{x_0}(\Omega)}}_{\text{on } \mathbb{R}^N} := \sum_{|\alpha|\le k} \frac{1}{\alpha!} D^\alpha f(x_0) D^\alpha g(x_0)$$

Let also $e_\alpha(x) := \frac{1}{\sqrt{\alpha!}}(x-x_0)^\alpha$. Then the $\{e_\alpha\}_{|\alpha|\le k}$ are an orthonormal basis of $\mathcal{P}_k(\Omega) \approx \mathbb{R}^N \approx J^k_{x_0}(\Omega)$ with respect to $\langle -,-\rangle_{J^k_{x_0}(\Omega)}$. Moreover,

$$\langle f, e_\beta\rangle_{J^k_{x_0}(\Omega)} = \sum_{|\alpha|\le k} \frac{1}{\alpha!} D^\alpha f(x_0) \cdot D^\alpha e_\beta(x_0) = \frac{1}{\beta!} D^\beta f(x_0) \cdot \sqrt{\beta!} = \frac{D^\beta f(x_0)}{\sqrt{\beta!}}$$

and thus

$$T^k_{x_0} f(x) = \sum_{|\alpha|\le k} \frac{D^\alpha f(x_0)}{\alpha!}(x-x_0)^\alpha = \sum_{|\alpha|\le k} \langle f, e_\alpha\rangle_{J^k_{x_0}(\Omega)} \cdot e_\alpha(x) \quad .$$

Taylor's Theorem Revisited: Let $f \in C^k(\Omega)$, and consider the subspace $\mathcal{P}^k(\Omega) \approx J^{x_0}_k(\Omega)$ as part of the direct sum decomposition

$$C^k(\Omega) = \mathcal{P}^k(\Omega) \oplus \mathcal{I}^{k+1}_{x_0}(\Omega)$$

Every $f \in C^k(\Omega)$ thus splits:

$$f(x) = T^k_{x_0} f(x) + R^k_{x_0}(x)$$

with $T^k_{x_0} f(x) = \sum_{|\alpha|\le k} \langle f, e_\alpha\rangle_{J^k_{x_0}(\Omega)} \cdot e_\alpha(x)$ as above, and $R^k_{x_0}(x) \in \mathcal{I}^{k+1}_{x_0}(\Omega)$.

Crucially, this splitting is unique, in that if one writes $f(x) = q(x) + H(x)$ for any other $q(x) \in \mathcal{P}^k(x)$, the corresponding remainder term $H = f - q$ will not vanish upto and including the k^{th} order at x_0: $H \notin \mathcal{I}^{k+1}_{x_0}(\Omega)$. Indeed, if $q \in \mathcal{P}^k(x)$, $q \neq T^k_{x_0} f$, then there must exist some α_0 with $|\alpha_0| \le k$ s.t.

$$D^{\alpha_0} q(x_0) \neq D^{\alpha_0}(T^k_{x_0} f)(x_0)$$

by the remark in Defn B. Thus,

$$D^{\alpha_0} H(x_0) = D^{\alpha_0}(f-q)(x_0) = D^{\alpha_0} f(x_0) - D^{\alpha_0} q(x_0) = D^{\alpha_0} T^k_{x_0}(x_0) - D^{\alpha_0} q(x_0) \neq 0.$$

(In fact, this observation is what proves that the sum of the subspaces $\mathcal{P}^k(\Omega)$ and $\mathcal{I}^{k+1}_{x_0}(\Omega)$ is a direct sum).

As such, we have a unique way to split up $f(x)$ into a degree $\leq k$ polynomial part, and a high order error term part. This this is the precise reason why we make our specific definition of $T^k_{x_0} f \in \mathcal{P}^k(\Omega)$.

Taylor's Theorem is thus a statement about the behavior of this high order error term $R^k_{x_0} f(x) \in \mathcal{I}^{k+1}_{x_0}(\Omega)$. In effect, Taylor's Theorem boils down to the statement that any C^k function $R \in \mathcal{I}^{k+1}_{x_0}(\Omega)$ satisfies the estimate

$$\lim_{x \to x_0} \frac{R(x)}{|x-x_0|^k} = 0,$$

or in other words, that $\mathcal{I}^{k+1}_{x_0}(\Omega) \subseteq o(|x-x_0|^k)$.

This leads us to a very nice rephrasing of Taylor's Theorem:

> A function $f: \Omega \to \mathbb{R}$ is C^k at x_0 iff $\exists\, p(x) \in \mathcal{P}^k(\Omega)$ such that $f(x) - p(x) \in \mathcal{I}^{k+1}_{x_0}(\Omega) \subseteq o(|x-x_0|^k)$. In this case, $p(x) = T^k_{x_0} f(x)$ is unique!

(‡) A Background Remark on Exact Sequences of Vector Spaces

Def: Let U, V, W be vector spaces, and suppose $S: V \to W$ is a surjective linear map and that $T: U \to V$ is an injective linear map. If $\ker(S) = \operatorname{im}(T)$ as subspaces of V, then we call the following diagram

$$0 \longrightarrow U \xrightarrow{T} V \xrightarrow{S} W \longrightarrow 0$$

a short exact sequence (exact just means that the kernel of a map in the sequence is exactly the image of the previous map).

Fact: For simplicity, assume that $U, W \subseteq V$ are already subspaces of V. Given such a setup, it follows that $V = U \oplus W$, meaning that every $v \in V$ can be uniquely written as $v = u + w$ for $u \in U$ and $v \in W$.

Application II: Critical Points of Functions

Let $U \subseteq \mathbb{R}^n$ be open.

Let $f: U \to \mathbb{R}$ be C^1. In applications, f may represent a quantity of interest that we would like to **minimize** or **maximize**. For instance, $f(x)$ could represent the cost of producing a part given parameters $(x^1, \dots, x^n)$. In general, finding such **extrema** $a \in U$ at which $f(x)$ is as big or small as possible (i.e. $f(x) \leq f(a)$ or $f(x) \geq f(a)$, resp., for all other choices of $x \in U$) is the subject of **optimization** and the **calculus of variations**, and is no small task. Here, however, we can already say a lot given the C^1 assumption on f.

Proposition 3.15: Suppose $f: U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$ is C^1, and that $a \in \overline{U}$ is an extremum. Then either (1) $a \in \partial U$ or (2) $a \in U$ and $Df(a) = 0$.

Proof: Since U is open, $\overline{U} = U \cup \partial U$, so indeed one of the two choices occurs.

Suppose $a \in U$, and let $\varepsilon > 0$ be so small that

$$a + te_i \in U \quad \text{for each } i = 1, \dots, n \text{ and all } t \in (-\varepsilon, \varepsilon).$$

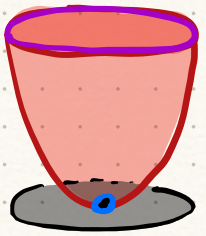
Since a is an extremum for $f(-)$ on U, all the single variable functions

$$t \mapsto f(a^1, \dots, a^i + t, \dots, a^n) \text{ have extrema at } t = 0, \text{ hence}$$

$$D_i f(a) = \frac{d}{dt}\Big|_{t=0} f(a^1, \dots, a^i + t, \dots, a^n) = 0.$$

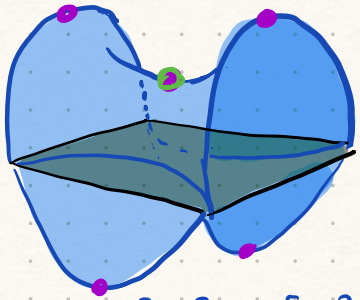
Definition 3.16: Any point $a \in \Omega$ at which $Df(a) = 0$ is called a **stationary**, or **critical point**, of $f: \Omega \to \mathbb{R}$.

Now, as you might recall from single variable calculus, **not every stationary point is an extremum!** Indeed, f might have an inflection point where $f'(a) = 0$, which is neither a local minimum, nor a local maximum. In the multivariate situation, the same is true, and, fascinatingly, there is even a great diversity in what sort of "inflection points" can occur. For instance, consider the following landscapes:



- maxima on Ω are not stationary
- minimum is stationary

$z = x^2 + y^2$ on $B_1(0)$



- Maxima and minima are not stationary
- The only stationary point is a "saddle point".

$z = x^2 - y^2$ on $[-1,1]\times[-1,1]$

Let's investigate this further: Fix a stationary point $a \in U$ of $f: U \to \mathbb{R}$, and assume $f \in C^2$, so that Taylor's Theorem tells us

$$f(x) - f(a) = Df(a)\cdot(x-a) + \tfrac{1}{2}(x-a)^t \cdot D^2 f(a)\cdot(x-a) + R_2(x)$$
$$= \tfrac{1}{2}\sum_{i,j=1}^{n} D_{ij} f(a)(x^i - a^i)(x^j - a^j) + R_2(x).$$

Thus, the sign of $f(x) - f(a)$, which tells us whether $f(x)$ is above or below $f(a)$, is controlled by higher order terms of the Taylor Series, starting with the Hessian terms. In particular, if $h \in \mathbb{R}^n$, $|h| = 1$, is a direction such that if $x = a + th$, $|t| << 1$, has

$$(x-a)^t \cdot D^2 f(a)\cdot(x-a) = t^2 (h^t \cdot D^2 f(a)\cdot h) > (<)\ 0,$$

then for small $|t| << 1$ we have $f(x) > (<)\ f(a)$, since

$$\frac{f(a+th) - f(a)}{t^2} = \underbrace{\tfrac{1}{2}(h^t \cdot D^2 f(a)\cdot h)}_{\text{has a definite sign}} + \underbrace{\frac{R_2(a+th)}{t^2}}_{\to 0}.$$

Thus, if $D^2 f(a)$ is such that $h^t \cdot D^2 f(a)\cdot h > 0$ (or < 0) for every $h \in \mathbb{R}^n$, $|h| = 1$, then we know for sure that a is a local minimum (or maximum).

If there are different such h which give opposite signs in this expression, then we can be sure that the point a is some sort of saddle point. The existence of directions h with $h^t \cdot D^2 f(a)\cdot h = 0$ complicates matters somewhat, and forces one to then consider higher order derivatives too.

Definition 3.17: A mapping $A(\cdot,\cdot): \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ which is a linear map in each argument is called a **bilinear map**. In particular, if $A \in M^{n\times n}(\mathbb{R})$ is a $(n\times n)$-matrix, the mapping $(x,y) \in \mathbb{R}^n \times \mathbb{R}^n \longmapsto x^t \cdot Ay \in \mathbb{R}$ is a bilinear mapping.

Given a bilinear map A, one can define a **quadratic form** $Q: \mathbb{R}^n \to \mathbb{R}$ by setting $Q(v) := A(v,v)$. As a particularly relevant example, if $A = D^2f(a)$ is the Hessian matrix of f, then the 2nd order term of the Taylor expansion of f is encoded as a quadratic form: If $h = x - a$,

$$\frac{1}{2}\sum_{i,j=1}^{n} D_{ij}f(x_0)(x^i - a^i)(x^j - a^j) = \frac{1}{2} h^t \cdot D^2f(x_0) \cdot h =: Q(h).$$

We say a quadratic form / the matrix with defines it is:

- positive (semi) definite iff $Q(h) >^{(\geq)} 0 \quad \forall h \in \mathbb{R}^n \setminus \{0\}$.
- negative (semi) definite iff $Q(h) <^{(\leq)} 0 \quad \forall h \in \mathbb{R}^n \setminus \{0\}$.

Therefore, to summarize the above discussion, at a stationary point $a \in U$ of f:

- if $D^2f(a)$ is positive definite, then a is a local minimum of f
- if $D^2f(a)$ is negative definite, then a is a local maximum of f

Thus, a multivariate second derivative test must give criteria for discerning when $D^2f(a)$ is positive/negative definite (or neither). Here is (without proof) one such test:

Theorem 3.18: Let $a \in U$ be a stationary point of $f: U \subseteq \mathbb{R}^n \to \mathbb{R}$ which is at least C^2. Let $d_1, d_2, \ldots, d_n$ be the **principal minors** of $D^2f(a)$:

$$d_1 = D_{11}f(a), \quad d_2 = \det\begin{pmatrix} D_{11}f(a) & D_{12}f(a) \\ D_{21}f(a) & D_{22}f(a) \end{pmatrix} \quad \cdots \quad d_n = \det\begin{pmatrix} D_{11}f(a) & \cdots & D_{1n}f(a) \\ \vdots & & \vdots \\ D_{n1}f(a) & \cdots & D_{nn}f(a) \end{pmatrix}$$

If $d_n \equiv \det D^2f(a) \neq 0$, we call $a \in U$ a **non degenerate critical point**.

In this case:
- if $d_k > 0$ for each $i = 1, \ldots, n$, then $D^2f(a)$ is **positive definite**;
- if $d_{even} > 0$ and $d_{odd} < 0$, then $D^2f(a)$ is **negative definite**;
- otherwise, $D^2f(a)$ is neither, and $a \in U$ is a **saddle point** of f.

If $d_n = \det D^2f(a) = 0$, $a \in U$ is said to be **degenerate**, and other* tests are needed.

* not in this lecture!

Examples / Exercises: Find the maxima / minima / saddle points of the following:

1. $f(x) = x \sin x$ on $[-1, 1]$.
2. $f(x^1, \dots, x^n) = |x|^2$ on $B_1(0)$.
3. $f(x, y) = x^2 - y^2$ on $[-1, 1] \times [-1, 1]$

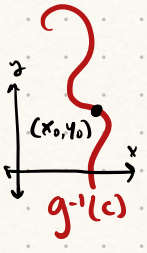
Application III: Constrained Optimization

General Problem of Interest: Let $f: U \subseteq \mathbb{R}^n \to \mathbb{R}$ be a C^1 function which we would like to minimize/maximize, with the added assumption that the independent variables $(x^1, \dots, x^n)$ must be constrained somewhow. In practice, such a situation might be the following:

- $f(x, y)$ represents the potential profit from producing x widgets, and y gadgets.
- An equation $g(x, y) = c$ (constant) describes the relationship between the amount x of widgets and amount y of gadgets that can be manufactured, eg, if a widget is 5 times more time consuming to produce than a gadget, and a manufacturer can produce only N many items total, then we want to maximize $f(x, y)$ subject to the constraint $g(x, y) = 5x + y = N$.

To get the geometric idea, imagine the 2D case above:

Problem: Find the extrema of a C^1 function $f: \mathbb{R}^2 \to \mathbb{R}$ subjected to the constraint that $g(x, y) = c$, for some C^1 function $g: \mathbb{R}^2 \to \mathbb{R}$.



Wherever $\nabla g(x_0, y_0) \neq 0$, it turns out that $g^{-1}(c)$ looks like a C^1 curve going through (x_0, y_0), so that $f|_{g^{-1}(c)}$ is just the restriction of f to a C^1 curve. If we find a way to locally parametrize this curve, say as $\gamma(t)$, then finding the extrema of $f|_{g^{-1}(c)}$ near to (x_0, y_0) is equivalent to finding the extrema of the single variable function $f \circ \gamma(t)$. This gives us a way to start searching for the extrema of $f(x)$ with x restricted to Σ. Points $(x_0, y_0) \in \Sigma$ where $\nabla g(x_0, y_0) = 0$ are then considered seperately "by hand."

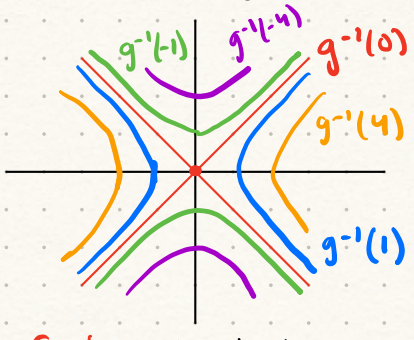
In general dimensions, a similar picture arises:

Definition 3.19: Let $g: \mathbb{R}^n \to \mathbb{R}$ be any map, and fix $c \in \mathbb{R}$. The set of points $g^{-1}(c) = \{x \in \mathbb{R}^n : g(x) = c\}$ is called a **level surface** of g, and whenever $\nabla g(x) \neq 0$ for $x \in g^{-1}(c)$ (assuming $g \in C^k$), $g^{-1}(c)$ forms a C^k smooth $(n-1)$-dimensional curved surface in $\mathbb{R}^n$ near x.

Once we prove the Implicit Function Theorem in Lecture 9, this last point about the regularity of the level surface will become clear, since that theorem will allow us to write the level surface, locally, as the graph of a C^k function!

Examples:

- If $g(x,y) = x^2+y^2$, $g^{-1}(1)$ is the unit circle in $\mathbb{R}^2$
- If $g(x,y,z) = x+y+z$, $g^{-1}(0)$ is a plane in $\mathbb{R}^3$ through 0.
- If $g_1(x,y,z,w) = x^2+y^2$ and $g_2(x,y,z,w) = z^2+w^2$, then $g_1^{-1}(\frac{1}{2}) \cap g_2^{-1}(\frac{1}{2})$ is a two dimensional torus inside of the three dimensional sphere $S^3 = \{x^2+y^2+z^2+w^2 = 1\}$, called the **Clifford Torus**.
- For $g(x,y) = x^2 - y^2$, $\nabla g(0,0) = 0$, and $g^{-1}(0)$ is not a smooth curve at the origin!



One of the most important facts about the level surfaces of a differentiable function, which will help us understand constrained optimization, is the following:

Theorem 3.20: Let $g: \mathbb{R}^n \to \mathbb{R}$ be a C^1 function and suppose $c \in \mathbb{R}$ is such that $g^{-1}(c)$ has the "regularity property" $\nabla g(x) \neq 0$ at $x \in g^{-1}(c)$. Then $\nabla g(x)$ is orthogonal to $g^{-1}(c)$ at x.

Remark/Definition: Said precisely, this means that the vector

$$\nabla g(x) = (D_1 g(x), \ldots, D_n g(x))$$

is orthogonal to every tangent vector of the surface $g^{-1}(c)$ at x: i.e. the vectors in $\mathbb{R}^n$ which lie in the tangent plane to the surface $g^{-1}(c)$ at x. However, despite the fact that it's "mathematically possible" to write $g^{-1}(c)$ as the graph of a function G near x (by the aforementioned Implicit Function Theorem), so that this plane is given by $\text{im}(DG)$ as before, it might be "practically impossible" to find G explicitly. Besides this practical roadblock, this definition of the tangent plane is not always "optimized" for use in certain arguments. Thus, it can be helpful on several levels to find alternative, equivalent characterizations.

An especially useful, intrinsic point of view (to the surface $\Sigma = g^{-1}(c)$) is the following: Let $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{R}^n$ be any C^1 curve with

- $\gamma(t) \in \Sigma = g^{-1}(c)$ for all $t \in (-\varepsilon, \varepsilon)$
- $\gamma(0) = x$.

Then $\dot{\gamma}(0)$ is said to be a tangent vector to Σ at x, and we write $\dot{\gamma}(0) = D\gamma_0(e_1) \in T_x\Sigma$. As a fact, one can show that $T_x\Sigma$ is indeed the union of all vectors so obtained, by considering all possible C^1 curves in Σ as above going through x.

In any case, via linear algebraic considerations it follows that $T_x\Sigma$ is an $n-1$ dim'l subspace of $\mathbb{R}^n$, and is the orthogonal complement of the 1-dim'l subspace $\text{span}\{\nabla g(x)\}$. Using this new characterization of $T_x\Sigma$, we can prove Theorem 3.20:

Proof: This result follows easily from the Chain rule and the above definition of tangent vectors to the surface $g^{-1}(c)$. Let $v \in T_x g^{-1}(c)$ and fix a C^1 curve $\gamma(t)$ so that • $\gamma(t) \in g^{-1}(c)$ • $\gamma(0) = x$ • $\dot{\gamma}(0) = v$.

We have: $$0 = \frac{d}{dt}\Big|_{t=0} g(\gamma(t)) = Dg_{\gamma(0)}(\dot{\gamma}(0)) = \langle \nabla g(x), v \rangle$$ ∎

Using this observation, we can now state our test for critical points of constrained functions:

Theorem 3.21 (Lagrange Multipliers) Let $\Omega \subseteq \mathbb{R}^n$ be a domain. Let $f: \Omega \to \mathbb{R}$ and $g: \Omega \to \mathbb{R}$ be C^1 functions, with the level set $g^{-1}(c)$ representing a constraint surface which we want to optimize f with respect to. That is, we seek the critical points of $f(x)$ when we restrict x to lie in $g^{-1}(c)$.

Then if $x_0 \in g^{-1}(c)$ is such that
- $\nabla g(x_0) \neq 0$
- x_0 is a critical point of $f|_{g^{-1}(c)}$

then there exists some scalar $\lambda \in \mathbb{R}$, called a Lagrange Multiplier, so that

$$\nabla f(x_0) = \lambda \nabla g(x_0).$$

Therefore, to optimize f subject to a constraint $g(x) = c$, we
- Solve the following system of $n+1$ equations in the $n+1$ unknowns $x^1, x^2, \dots, x^n, \lambda \in \mathbb{R}$:

$$\begin{cases} \nabla f(x) = \lambda \nabla g(x) \\ g(x) = c \end{cases}$$

- Find all points $x \in g^{-1}(c)$ where $\nabla g(x) = 0$. If Ω has boundary and $g^{-1}(c)$ meets $\partial\Omega$, consider also these points.

Together, all these points comprise the locations where $f|_S$ could have critical points and extrema (in particular maxima and minima).

Proof: Suppose $x_0 \in \text{int}(\Omega)$ is a point where $\nabla g(x_0) \neq 0$. Then $\Sigma := g^{-1}(c)$ is a C^1 surface (of dim $= n-1$) in an open neighborhood of x_0, and $\nabla g(x_0)$ is orthogonal to $T_{x_0}\Sigma$.

Now, for any C^1 curve $\gamma: (-\varepsilon, \varepsilon) \to \Sigma$ with $\gamma(0) = x_0$, the restricted C^1 function $F(t) = f(\gamma(t))$ must have a critical point at $t = 0$, since f has a critical point at $\gamma(0) = x_0$.

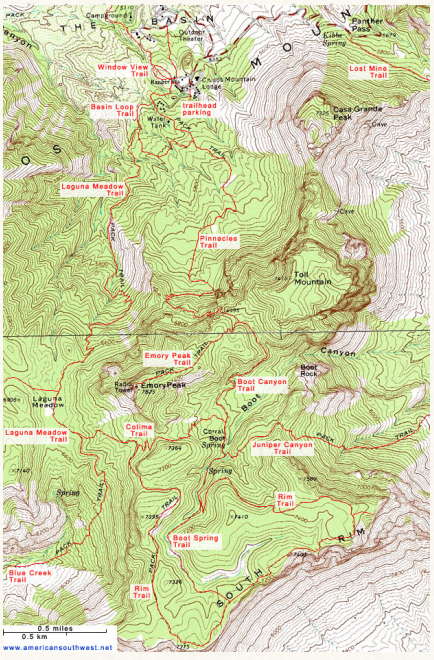
Thus, $$0 = \frac{d}{dt}\Big|_{t=0} F(t) = Df_{\gamma(0)}(\dot\gamma(0)) = \langle \nabla f(x_0), \dot\gamma(0)\rangle.$$
By arbitrariness of γ, it follows that $\nabla f(x_0)$ is orthogonal to $T_{x_0}\Sigma$. Then, since $T_{x_0}\Sigma$ is the orthogonal complement of $\operatorname{span}\{\nabla g(x_0)\}$, anything orthogonal to $T_{x_0}\Sigma$, such as $\nabla f(x_0)$, must be in $\operatorname{span}\{\nabla g(x_0)\}$.

Remark: Sometimes, one also encounters the situation of having multiple constraints $g_1(x) = c_1$, $g_2(x) = c_2$, ..., $g_m(x) = c_m$. If $m < n$, then provided
$$\operatorname{rank}\begin{bmatrix} \nabla g_1^1(x) & \cdots & \nabla g_m^1(x) \\ | & & | \end{bmatrix} = m \qquad \text{at } x,$$
it turns out (using the Implicit Function Theorem, as usual) that $\Sigma := g_1^{-1}(c_1) \cap \cdots \cap g_m^{-1}(c_m)$ is also a C^1 surface (as the graph of a certain C^1 function) in some small neighborhood of $x \in \Sigma$, and it has $\dim(\Sigma) = n - m$:

Thus finding stationary points (eg minima and maxima) of $f : \Omega \to \mathbb{R}$ subject to the constraint $g_i(x) = c_i$ for $i = 1, \ldots, m$, is effectively the same as above: at any regular point $x_0 \in \Sigma$ (i.e. x_0 satisfies the rank condition above, so that $\{\nabla g_i(x_0)\}$ is a linearly independent set), $T_{x_0}\Sigma$ is the orthogonal complement of $\operatorname{span}\{\nabla g_1(x_0), \ldots, \nabla g_m(x_0)\}$. For any C^1 curve $\gamma(t)$ in Σ with $\gamma(0) = x_0$, $f(\gamma(t))$ has a critical point at $t = 0$ if x_0 is a critical point of $f|_\Sigma$, hence $\nabla f(x_0) \in \operatorname{span}\{\nabla g_1(x), \ldots, \nabla g_m(x)\}$. That is, there exist $\lambda^1, \ldots, \lambda^m \in \mathbb{R}$ such that
$$\begin{cases} \nabla f(x_0) = \lambda^1 \nabla g_1(x_0) + \cdots + \lambda^m \nabla g_m(x_0) \\ g_i(x_0) = c_i \end{cases}$$

Here's a (hopefully) intuitive interpretation of Lagrange multipliers:

Intuition in 2D with 1-constraint: Let $f: \Omega \subseteq \mathbb{R}^2 \to \mathbb{R}$ be the height above sea level in Big Bend National Park, and let $g: \Omega \to \mathbb{R}$ be such that $g(x,y) = 0$ implicitly describes a hiking trail through the park.



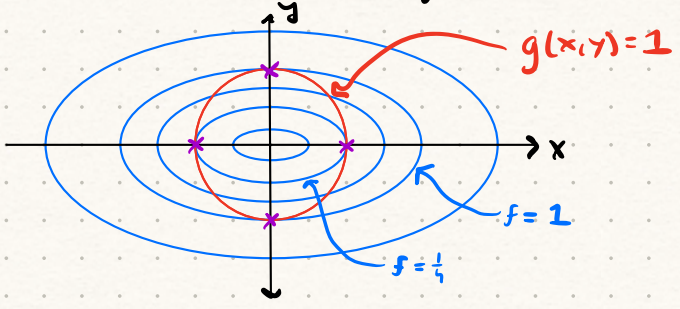
The topographic map at left shows trails in red (various constraint curves $g(x,y)=0$) and contour lines (points of constant elevation $f(x,y)=c$) in brown.

A moment's thought should convince you that the maximum point of elevation along a trail should occur at a point where the trail is tangent to a contour line (or else at the boundary of the park). If not, the trail would cross the contour line, so that you could immediately move to a contour line of greater elevation value. Ditto for general stationary points.

Let's do some examples:

(1) Find the extrema of $f(x,y) = \frac{1}{4}x^2 + y^2$ subject to the constraint $g(x,y) = x^2 + y^2 = 1$.

Geometrically:



Analytically: $\Omega = \mathbb{R}^2$ has no boundary, so we only need to find critical points of the constrained problem. Since $\nabla g(x,y) = (2x, 2y) = 0$ iff $(x,y) = 0$, which does not satisfy the constraint $g(x,y) = 1$, all possible extrema of the problem will appear as solutions of the system

$$\begin{cases} \left(\frac{1}{2}x, 2y\right) = \nabla f(x,y) = \lambda \nabla g(x,y) = \lambda(2x, 2y) \\ g(x,y) = x^2 + y^2 = 1 \end{cases}$$

Thus we see that $\begin{cases} \text{(i)}\ x = 4\lambda x \\ \text{(ii)}\ y = \lambda y \\ \text{(iii)}\ x^2+y^2 = 1 \end{cases}$

(i) $\Rightarrow x = 0$ or else $\lambda = \frac{1}{4}$

(ii) $\Rightarrow y = 0$ or else $\lambda = 1$

(iii) $\Rightarrow$: If $\underbrace{x=0,\ y=\pm 1}_{\lambda=1}$. If $\underbrace{y=0,\ x=\pm 1}_{\lambda=1/4}$.

Thus we have solutions $(0,\pm 1)$ and $(\pm 1, 0)$, and it is trivial to check that $(0,\pm 1)$ are maxima, $(\pm 1, 0)$ are minima.

An aside on what λ represents: Notice that the method of Lagrange multipliers transforms the task of finding critical points of $f(\vec{x})$ subject to the constraint $g(\vec{x}) = c$ into the task of finding the unconstrained critical points of the new function

$$L(x,\lambda) = f(x) - \lambda(g(x) - c)$$

on $\Omega \times \mathbb{R}$. Indeed,

$$0 = \nabla_{x,\lambda} L(x,\lambda) = \left(D_1 L, \dots, D_n L, D_\lambda L\right) = \left(\nabla f(x) - \lambda \nabla g(x),\ c - g(x)\right)$$

enforces the whole system of equations

$$(*) \begin{cases} \nabla f(x) = \lambda \nabla g(x) \\ g(x) = c \end{cases}.$$

Let's now think of c as another parameter:

$$L(x,\lambda,c) = f(x) - \lambda(g(x) - c)$$

Let $(x_0, \lambda_0, c_0) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}$ denote a critical point of L, and suppose (under mild assumptions on L as in the Implicit Function Theorem) that for $c \in \mathbb{R}$ near to c_0, the corresponding critical points $x(c)$ near $x(c_0) = x_0$ are parametrized by a C^1 curve, so that

1. $g(x(c)) = c$
2. $(x(c), \lambda(c), c)$ is a critical point of $L(x,\lambda,c)$.

If we define $V(c) = L(x(c), \lambda(c), c) = f(x(c))$

to be the critical value of f at this point, then we have that

$$\begin{aligned} V'(c_0) &= Df_{x(c_0)}(x'(c_0)) \\ &= \lambda(c_0) Dg_{x(c_0)}(x'(c_0)) \qquad \text{(Lagrange Multiplier)} \\ &= \lambda(c_0) \frac{d}{dc}\Big|_{c=c_0} g(x(c)) \\ &= \lambda(c_0) \frac{d}{dc}\Big|_{c=c_0} c \\ &= \lambda(c_0). \end{aligned}$$

Therefore, in the case where we can find a smooth path of critical points $x(c)$ for c near to c_0, we see that $\lambda(c_0)$ is the rate of change of the critical value $f(x(c_0))$ as we change c_0!

Remark: As we will see in the next lectures on the Inverse and Implicit Function Theorems, guaranteeing this path of critical points is not so difficult.

The idea: We seek to solve the equation

$$\begin{cases} L(x, \lambda, c): \mathbb{R}^n \times \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R} \\ \mathcal{F}(x, \lambda, c) = (\nabla_{(x,\lambda)} L)(x, \lambda, c) = 0 \end{cases}$$

for $(x, \lambda) \in \mathbb{R}^{n+1}$ in terms of the parameter $c \in \mathbb{R}$. We know that

$$\mathcal{F}(x_0, \lambda_0, c_0) = 0$$

Noting that $\mathcal{F}(x, \lambda, c)$ defines $(n+1)$-equations in $(n+2)$-unknowns, it seems plausible that we should be able to treat one variable as a parameter, and then have the other $(n+1)$ variables determined by the resulting $(n+1)$-eqtns. We'll explore how to do this next time!